Machine Learning Fundamentals

Original educational benchmark document for testing document retrieval and grounded question answering.

1. What Machine Learning Means

Machine learning is a field of computing in which a system improves its performance on a task by learning patterns from data. Instead of writing every decision as a fixed rule, a developer provides examples or observations from which a model can learn.

A dataset is a collection of examples used by a machine-learning workflow. A row is often called a record or instance, while a column is often called a feature. In a supervised-learning dataset, a target is the value the model is trained to predict.

A model is a mathematical representation learned from data. Training is the process of adjusting the model using examples. Inference is the later process of using the trained model to produce a prediction for new input.

Machine learning does not automatically make a system correct. Results depend on data quality, the choice of target, the suitability of the model, and how performance is measured.

2. Main Learning Setups

Supervised learning uses examples that contain both input features and known target values. Classification predicts a category, such as a message label. Regression predicts a numerical value, such as a delivery time.

Unsupervised learning works with data that does not include supplied target labels. Clustering groups examples according to similarities in their features. Dimensionality reduction represents data with fewer variables while attempting to retain useful structure.

Reinforcement learning studies an agent that interacts with an environment. The agent takes actions, receives rewards or penalties, and aims to learn a policy that performs well over time.

The learning setup should match the problem. For example, predicting a customer's numeric spending amount is a regression task, while assigning a support ticket to a department is a classification task.

3. Data Splitting and Generalization

A training set is used to fit model parameters. A validation set is used during development to compare alternatives or tune settings. A test set is held back until final evaluation to estimate performance on unseen examples.

Data leakage occurs when information unavailable at prediction time influences training or evaluation. Leakage can make reported performance look better than the system will perform in real use. Preprocessing steps should be learned from the training data and then applied consistently to validation and test data.

Overfitting occurs when a model learns details or noise specific to its training examples and performs poorly on new data. Underfitting occurs when a model is too limited to capture important patterns in the data.

Generalization is the ability of a trained model to perform well on new examples drawn from the intended use setting. A reliable evaluation process keeps test data separate from decisions made during model development.